ER Model and Relational Model

Database Management Systems - Unit 1 - Data modelling fundamentals

Why databases?

A database management system (DBMS) stores shared data in a structured way and gives applications safe, concurrent access to it. Compared with plain files, a DBMS reduces redundancy, enforces integrity rules, supports transactions and separates how data is stored from how it is used (data independence).

Entity-Relationship (ER) model

The ER model describes a system as entities, attributes and relationships before any tables are designed.

- Entity: a distinguishable thing, e.g. Student, Course. A weak entity cannot be identified without its owner entity.
- Attribute: a property of an entity. Types: simple, composite (Name = First + Last), multivalued (Phone), derived (Age from DateOfBirth).
- Relationship: an association between entities, e.g. Student ENROLLS_IN Course.
- Cardinality: 1:1, 1:N, M:N. Participation: total (every instance must take part) or partial.

> *Worked example: a Student may enrol in many Courses and a Course has many Students, so ENROLLS_IN is M:N. It becomes its own table holding both keys.*

Mapping ER to relational tables

- Each strong entity becomes a table; the key attribute becomes the primary key.
- A 1:N relationship adds the key of the '1' side as a foreign key on the 'N' side.
- An M:N relationship becomes a new table whose primary key combines both foreign keys.
- A multivalued attribute becomes a separate table referencing the entity.

Keys

- Super key: any set of attributes that uniquely identifies a row.
- Candidate key: a minimal super key. Primary key: the candidate key chosen as the main identifier.
- Foreign key: attribute(s) referencing a primary key in another table - this enforces referential integrity.

```
CREATE TABLE Student (
  roll_no   INT PRIMARY KEY,
  name      VARCHAR(60) NOT NULL,
  email     VARCHAR(80) UNIQUE
);
CREATE TABLE Enrollment (
  roll_no   INT REFERENCES Student(roll_no),
```

```
  course_id INT REFERENCES Course(course_id),
  PRIMARY KEY (roll_no, course_id)
);
```

Relational algebra in one minute

- Selection (sigma) picks rows that satisfy a condition.
- Projection (pi) picks columns and removes duplicates.
- Union, intersection and difference need union-compatible relations.
- Join combines related rows; natural join matches on equal common attributes.